Causal Obstruction–Response Theory of Transformer Decisions

A Local-to-Global Framework for Decision Formation, Repair, and Stability

Visya Gregoriy
visyagregoriy@gmail.com

Abstract

Modern language models do not make decisions in a single place. A factual answer, multiple-choice selection, next-token commitment, or tool-use action emerges from distributed computations across tokens, layers, attention heads, MLPs, residual streams, and learned feature directions. Yet many tools for analyzing these decisions focus either on final confidence scores, such as entropy and logit margin, or on locating individual components whose intervention changes the output. These views are useful but incomplete: they do not directly measure whether the model's distributed internal evidence is mutually compatible. We propose *Causal Obstruction–Response Theory*, a local-to-global framework for Transformer decisions. For a candidate contrast $c = (a, b)$, we represent local decision evidence as germs over a computation complex, define compatibility maps between internal sites, and measure failure of local evidence to glue into a stable global decision structure using a corrected obstruction energy. The obstruction is defined metrically as distance to the space of compatible global sections and computed through a weighted sheaf-Laplacian residual, avoiding the invalid construction $[ds] \in H^1$ for a globally defined cochain. We then couple obstruction to a causal response rule that predicts which activation patch, ablation, residual edit, or feature intervention should reduce obstruction and change the candidate margin. The central empirical loop is falsifiable: obstruction predicts intervention sites; predicted interventions change obstruction; obstruction change predicts logit-margin change; and margin change flips, repairs, or stabilizes decisions. This paper develops the conceptual, mathematical, algorithmic, and experimental blueprint for testing local-to-global decision mechanics in Transformers.

1 Introduction

Modern language models do not make decisions in a single place. A factual answer, multiple-choice selection, next-token commitment, tool-use action, or reasoning branch emerges from distributed computations across token positions, attention heads, MLPs, residual streams, normalization operations, and learned feature directions. Mechanistic interpretability has made this distribution visible: activation patching, causal tracing, attribution patching, circuit discovery, sparse autoencoders, and attribution graphs all show that behaviorally relevant information is spread across many internal sites rather than stored in one neuron or one scalar confidence value.

Yet our formal language for describing decisions often remains poorer than the computations we study. We typically ask which component is important, which feature is active, or how large the final logit margin is. These are useful questions, but they do not directly ask whether the model's distributed internal evidence coheres into a stable decision. A model may answer correctly because many local computations support the same conclusion. It may answer correctly but fragilely because a few late-layer features overpower unresolved conflict elsewhere. It may answer incorrectly because local evidence supporting different answers fails to integrate. It may hallucinate with high confidence because the final distribution is sharp even though intermediate computations remain mutually incompatible.

We propose a local-to-global view of Transformer decisions. For a fixed prompt and candidate answer contrast, we model internal evidence as a collection of local decision germs assigned to computational sites.

These sites may be residual-stream vectors, attention-head outputs, MLP outputs, sparse-autoencoder features, or higher-level feature nodes. Edges between sites encode compatibility: how evidence at one site should transport to another if the model's computation is internally coherent with respect to the candidate contrast. A stable decision is then not identified with a literal single global neuron, nor with confidence alone, but with the degree to which these local germs can be glued into a compatible global decision structure.

The key object is an obstruction energy. In exact sheaf theory, one might be tempted to define an obstruction as a cohomology class $[ds] \in H^1$. That is not the right construction here: if s is a globally defined 0-cochain, then ds is a coboundary and its cohomology class is zero. Instead, we define obstruction metrically. Given a candidate-specific decision sheaf F_c over a computation complex K_x, and a local evidence state $s_c(x)$, the obstruction is the distance from $s_c(x)$ to the space of compatible global sections:

$$\mathcal{D}_c(x) = \min_{g \in H^0(K_x; F_c)} \|s_c(x) - g\|^2.$$

For computation, we use the equivalent finite-dimensional residual energy:

$$\mathcal{O}_c(x) = \|d_c s_c(x)\|^2 = \sum_{e:u \to v} w_e \|\rho^c_{u \to v} s_c(u) - s_c(v)\|^2.$$

This measures not whether the model is confident, but whether candidate-relevant internal evidence is compatible across sites.

Obstruction alone is not enough. A high obstruction score is scientifically useful only if it predicts causal interventions. We therefore couple obstruction to response theory. Given an activation patch, ablation, residual edit, steering vector, or feature intervention, we estimate its local effect on both obstruction and candidate margin. The resulting obstruction-response score ranks internal sites by predicted obstruction reduction, predicted margin change, and off-target KL/Fisher cost. The framework is falsifiable through a closed loop: compute obstruction, predict intervention sites, intervene, measure obstruction change, measure logit-margin change, and observe whether the model's decision flips, repairs, or stabilizes.

Contributions. This paper contributes: (i) a local-to-global theory of Transformer decision formation; (ii) a candidate-specific decision-sheaf formalism for internal evidence; (iii) a corrected obstruction energy based on distance to global compatibility rather than the invalid cohomology class of an exact coboundary; (iv) a causal response rule for predicting intervention-effective sites; (v) an experimental protocol for testing repair, flip, and stabilization; and (vi) a limitation-aware account of non-identifiability, basis dependence, and overclaiming risks.

2 The Paper's Goal and Thesis

The goal of this paper is to make local-to-global consistency a testable scientific theory of Transformer decision formation. In one sentence: *we aim to show that candidate decisions in Transformers can be modeled as distributed compatibility events, and that a computable obstruction-response functional predicts causal interventions that repair, flip, or stabilize those decisions.*

The central thesis is:

> *AI decisions are local-to-global consistency events: for a candidate decision, distributed local evidence across internal computational sites must become sufficiently compatible to support a stable global commitment.*

The reviewer-safe version is:

For a fixed input and candidate contrast, a Transformer induces local decision evidence across computational sites. Stable decisions tend to exhibit low candidate-specific incompatibility relative to their output margin, while brittle, wrong, ambiguous, or overconfident decisions may exhibit measurable obstruction. A causal response rule can test this hypothesis by predicting which internal interventions should reduce obstruction and alter the final decision.

This is not a claim that all cognition is literally a global section. It is a modeling principle with a falsifiable instantiation.

3 Main Object: The Obstruction–Response Functional

The central object of the paper is the **obstruction-response functional**:

$$\mathcal{R}_c(u, \delta; x) = \alpha[-\widehat{\Delta\mathcal{O}}_{cu,\delta}(x)] + \beta\widehat{\Delta m}_{cu,\delta}(x) - \gamma\widehat{\Delta\mathrm{Cost}}_{u,\delta}(x). \tag{1}$$

Everything else serves this object. The computation complex defines where local evidence lives. The decision sheaf defines compatibility. The obstruction energy measures incompatibility. The response rule predicts how interventions change obstruction and margin. Decision boundaries explain flips. Geometry and Fisher penalties define intervention cost. Experiments test whether the functional predicts actual causal effects.

4 What This Paper Is and Is Not

Table 1: Positioning of the paper.

This paper is	**This paper is not**
A theory of distributed decision formation	Just activation patching with new notation
A mathematical framework for local-to-global decision evidence	Just sheaf theory applied decoratively
A causal intervention method for repairing, flipping, and stabilizing decisions	Just topology, QFT, or differential geometry
An empirical test of local-to-global consistency in Transformers	Just a hallucination detector
A framework for measuring incompatibility beyond confidence and margin	A benchmark-only paper
A foundation for future AI decision science	A claim that all AI cognition is literally a global section
A limitation-aware method for intervention-effective structures	A claim to identify the unique true circuit

5 Formal Framework

5.1 Candidate Contrasts

Let M_θ be a frozen Transformer and x an input prompt. For candidate answers a and b, define the candidate contrast

$$c = (a, b). \tag{2}$$

The candidate margin is

$$m_c(x) = z_a(x) - z_b(x), \tag{3}$$

where z_a, z_b are logits or task scores.

5.2 Computation Complex

Definition 1 (Computation complex)**.** *For an input x, the computation complex is a finite directed complex*

$$K_x = (V_x, E_x), \tag{4}$$

where vertices $v \in V_x$ are internal computational sites and edges $e : u \to v$ encode computational, causal, or learned compatibility relations.

A vertex may be indexed as

$$v = (\ell, t, q), \tag{5}$$

where ℓ is layer, t is token position, and q is a component type such as residual stream, attention-head output, MLP output, sparse-autoencoder feature block, or transcoder feature.

5.3 Local Decision Germs

Definition 2 (Candidate-specific stalk)**.** *For a candidate contrast c, each site $v \in V_x$ is assigned a finite-dimensional evidence space*

$$F_c(v) \cong \mathbb{R}^{r_v}. \tag{6}$$

Let $\phi_v(x) \in \mathbb{R}^{d_v}$ be the activation or feature vector at site v. A local decision germ is

$$s_c(v; x) = P_v^c \phi_v(x) \in F_c(v), \tag{7}$$

where P_v^c is a candidate-specific projection, probe, feature map, or logit-lens contrast.

The full local evidence state is a 0-cochain:

$$s_c(x) \in C^0(K_x; F_c) = \prod_{v \in V_x} F_c(v). \tag{8}$$

5.4 Compatibility Maps

For every edge $e : u \to v$, define a candidate-specific transport map

$$\rho_{u \to v}^c : F_c(u) \to F_c(v). \tag{9}$$

In experiments, these maps may be learned by ridge regression on clean examples:

$$\rho_{u \to v}^c = \arg\min_R \sum_{x_i \in \mathcal{T}_{\text{clean}}} \|R s_c(u; x_i) - s_c(v; x_i)\|^2 + \lambda \|R\|_F^2. \tag{10}$$

They may also be approximated through local Jacobians:

$$\rho_{u \to v}^c \approx P_v^c \frac{\partial \phi_v}{\partial \phi_u} (P_u^c)^\dagger. \tag{11}$$

5.5 Coboundary and Global Sections

Define the coboundary operator $d_c : C^0(K_x; F_c) \to C^1(K_x; F_c)$ by

$$(d_c s_c)(e : u \to v) = \rho^c_{u\to v} s_c(u) - s_c(v). \tag{12}$$

The compatible global decision sections are

$$H^0(K_x; F_c) = \ker d_c = \{g \in C^0(K_x; F_c) : d_c g = 0\}. \tag{13}$$

5.6 Corrected Obstruction Energy

A tempting but incorrect construction is to call $[d_c s_c] \in H^1(K_x; F_c)$ the obstruction. This is invalid in the present setting because s_c is a globally defined 0-cochain, so $d_c s_c$ is exact and its cohomology class is zero.

Instead, the obstruction is metric.

Definition 3 (Distance-to-global-section obstruction)**.** *The obstruction of $s_c(x)$ is*

$$\mathcal{D}_c(x) = \min_{g \in H^0(K_x;F_c)} \|s_c(x) - g\|^2_M. \tag{14}$$

Definition 4 (Laplacian residual obstruction)**.** *The computable obstruction energy is*

$$\mathcal{O}_c(x) = \|d_c s_c(x)\|^2_W = \sum_{e:u\to v} w_e \|\rho^c_{u\to v} s_c(u; x) - s_c(v; x)\|^2. \tag{15}$$

Let $L_c = d_c^* W d_c$. Then

$$\mathcal{O}_c(x) = s_c(x)^\top L_c s_c(x). \tag{16}$$

Normalize obstruction by evidence magnitude:

$$\widetilde{\mathcal{O}_c}(x) = \frac{\mathcal{O}_c(x)}{\|s_c(x)\|^2_M + \epsilon}. \tag{17}$$

Define fragility by comparing obstruction to margin:

$$\mathrm{Frag}_c(x) = \frac{\widetilde{\mathcal{O}_c}(x)}{|m_c(x)| + \epsilon}. \tag{18}$$

6 Obstruction–Response Intervention Theory

Let h_u be the activation at site u, and let an intervention δ_u be injected by

$$h \mapsto h + B_u \delta_u. \tag{19}$$

The linearized margin response is

$$\Delta m_c \approx J^m_{c,u} \delta_u, \qquad J^m_{c,u} = \frac{\partial m_c}{\partial h_u}. \tag{20}$$

The obstruction response is

$$\Delta \mathcal{O}_c \approx \nabla_{h_u} \mathcal{O}_c^\top \delta_u + \frac{1}{2} \delta_u^\top H^{\mathcal{O}}_u \delta_u. \tag{21}$$

To penalize collateral damage, define an off-target KL or Fisher cost:

$$\Delta\mathrm{Cost}_u \approx \frac{1}{2}\delta_u^\top F_u \delta_u, \tag{22}$$

where

$$F_u = \mathbb{E}_{y\sim p_\theta(\cdot|x)}\left[\nabla_{h_u}\log p_\theta(y \mid x)\nabla_{h_u}\log p_\theta(y \mid x)^\top\right]. \tag{23}$$

The intervention objective is

$$\delta_u^\star = \arg\min_\delta \left[\mathcal{O}_c(h + B_u\delta) - \beta m_c(h + B_u\delta) + \frac{\mu}{2}\delta^\top F_u\delta + \frac{\lambda}{2}\|\delta\|^2\right]. \tag{24}$$

Under a local quadratic approximation,

$$\delta_u^\star = -\left(H_u^O + \mu F_u + \lambda I\right)^{-1}\left(\nabla_{h_u}\mathcal{O}_c - \beta(J_{c,u}^m)^\top\right). \tag{25}$$

For practical ranking, define

$$S_u = \alpha[-\widehat{\Delta\mathcal{O}_{cu}}] + \beta\widehat{\Delta m_{cu}} - \gamma\widehat{\Delta\mathrm{Cost}}_u. \tag{26}$$

7 Algorithm

Algorithm 1 Obstruction–Response Intervention for Transformer Decisions

Require: Model M_θ, prompt x, candidates a, b, sites V, edges E, learned maps ρ_e^c, intervention budget k
Ensure: Obstruction score, fragility score, ranked intervention sites, predicted and observed decision changes

Define candidate contrast $c = (a, b)$
Run $M_\theta(x)$ and cache activations $\{\phi_v(x)\}_{v\in V}$
Compute local germs $s_c(v; x) = P_v^c\phi_v(x)$
Compute margin $m_c(x) = z_a(x) - z_b(x)$
for each edge $e : u \to v$ **do**
    Compute residual $r_e = \rho_e^c s_c(u; x) - s_c(v; x)$
end for
Compute obstruction $\mathcal{O}_c(x) = \sum_e w_e\|r_e\|^2$
Compute normalized obstruction $\widetilde{\mathcal{O}_c}(x)$ and fragility $\mathrm{Frag}_c(x)$
for each intervention site u **do**
    Choose intervention direction δ_u by patching, ablation, residual edit, or feature steering
    Estimate $\widehat{\Delta m_{cu}} = J_{c,u}^m\delta_u$
    Estimate $\widehat{\Delta\mathcal{O}_{cu}} = \nabla_{h_u}\mathcal{O}_c^\top\delta_u + \frac{1}{2}\delta_u^\top H_u^O\delta_u$
    Estimate off-target cost $\widehat{\Delta\mathrm{Cost}}_u$
    Score $S_u = \alpha[-\widehat{\Delta\mathcal{O}_{cu}}] + \beta\widehat{\Delta m_{cu}} - \gamma\widehat{\Delta\mathrm{Cost}}_u$
end for
Patch, ablate, or steer the top k sites by S_u
Re-run model and measure actual $\Delta\mathcal{O}_c$, Δm_c, KL, and decision outcome
return scores, rankings, predicted changes, observed changes

8 Decision Boundaries, Flips, and Singular Regimes

A candidate decision flips when an intervention moves the model state across the boundary

$$m_c(h) = 0. \tag{27}$$

Under local linearization, frozen attention, or a fixed piecewise-linear activation region, the margin is approximately affine:

$$m_c(h) \approx A_c h + b_c. \tag{28}$$

Thus, locally, the decision boundary is

$$\{h : A_c h + b_c = 0\}. \tag{29}$$

Singular regimes arise when the response map loses rank, when the projection onto $H^0(K_x; F_c)$ becomes ill-conditioned, or when small interventions cause discontinuous changes in active computational pathways. Algebraic, tropical, or singularity-theoretic language should be used only under such explicit approximations.

9 Theorem Targets

Theorem 1 (Energy–distance equivalence)**.** *Let $d_c : C^0 \to C^1$ be the coboundary of a finite Hilbert sheaf with positive edge weights W, and let $L_c = d_c^* W d_c$. Let $P_{\ker d_c}$ be the M-orthogonal projection onto $H^0 = \ker d_c$. Then, on $(\ker d_c)^\perp$, there exist spectral constants $0 < \lambda_{\min}^+ \le \lambda_{\max}$ such that*

$$\lambda_{\min}^+ \|(I - P_{\ker d_c})s\|_M^2 \le s^\top L_c s \le \lambda_{\max} \|(I - P_{\ker d_c})s\|_M^2. \tag{30}$$

Proposition 2 (Minimum-energy repair under linear response)**.** *Assume $\mathcal{O}_c$ is locally quadratic, m_c is locally linear, and the off-target cost is quadratic with positive semidefinite Fisher matrix F_u. Then the local optimal repair intervention is*

$$\delta_u^\star = -\left(H_u^O + \mu F_u + \lambda I\right)^{-1} \left(\nabla_{h_u} \mathcal{O}_c - \beta (J_{c,u}^m)^\top\right). \tag{31}$$

Proposition 3 (Obstruction–margin stability)**.** *Suppose margin response is locally Lipschitz under perturbations of norm at most r. If*

$$|m_c(x)| > \sup_{\|\delta\| \le r} |J_m \delta| + \tau \mathcal{O}_c(x), \tag{32}$$

then the candidate ordering is stable under the allowed perturbation class.

Proposition 4 (Non-identifiability)**.** *In redundant or superposed representations, multiple low-overlap compatibility structures may induce similar obstruction-response predictions for the same candidate contrast.*

Proposition 5 (Local decision boundary)**.** *Under frozen attention and a fixed piecewise-linear activation region, a candidate margin boundary is locally affine:*

$$\{h : m_c(h) = 0\} = \{h : A_c h + b_c = 0\}. \tag{33}$$

Proposition 6 (Relative extension obstruction)**.** *Let $U \subset K$ be a subcomplex and let s_U be a fixed partial section. A true cohomological obstruction to extending s_U to a global section can live in relative cohomology:*

$$\mathrm{Obs}(s_U) \in H^1(K, U; F). \tag{34}$$

10 Experimental Blueprint

The experiments should tell a cumulative story rather than merely demonstrate one trick.

10.1 Experiment 1: IOI and Known Circuits

Question. Does obstruction-response predict which residual-stream or component sites causally repair corrupted IOI decisions?

Models and tasks. Use GPT-2 small with TransformerLens on IOI clean/corrupt prompts.

Metrics. Top-k repair rate, predicted-vs-observed Δm_c, predicted-vs-observed $\Delta \mathcal{O}_c$, AUC for effective sites, and off-target KL.

Supporting result. Obstruction-response rankings outperform random, activation norm, gradient saliency, attribution patching, and simple patching heuristics.

Weakening result. The score is no better than attribution patching, or obstruction does not change when patches repair the answer.

10.2 Experiment 2: Factual Recall

Question. Does obstruction-response generalize from synthetic circuits to factual decisions?

Models and tasks. Use GPT-2 medium, Pythia models, and factual recall datasets such as CounterFact or zsRE-style prompts.

Metrics. Repair rate for corrupted or counterfactual prompts, margin change, obstruction reduction, and KL collateral damage.

Supporting result. Predicted interventions repair factual answers with fewer sites and lower KL than baselines.

10.3 Experiment 3: Multiple-Choice and Reasoning Decisions

Question. Does the framework apply to candidate decisions beyond next-token factual recall?

Models and tasks. Use ARC, MMLU subsets, or HellaSwag with candidate contrasts $c = (\text{correct}, \text{distractor})$.

Metrics. Candidate-margin improvement, answer flips, calibration of flip probability, and transfer across paraphrases.

10.4 Experiment 4: Ambiguity and Conflicting Evidence

Question. Do ambiguous prompts produce high obstruction relative to margin?

Models and tasks. Construct prompts with misleading context, conflicting cues, or late corrections.

Metrics. Fragility, paraphrase instability, repair by targeted intervention, and obstruction change after disambiguating context.

10.5 Experiment 5: Hallucination and Overconfidence

Question. Can obstruction identify overconfident but internally incompatible decisions?

Models and tasks. Use TruthfulQA-style prompts or factual prompts with uncertain premises.

Metrics. Obstruction at fixed confidence, semantic-entropy comparison, and intervention sensitivity.

10.6 Experiment 6: SAE-Feature Version

Question. Does feature-space obstruction improve over residual-space obstruction?

Models and tasks. Use open-weight models with available sparse autoencoders.

Metrics. Intervention efficiency, interpretability of selected sites, and robustness to feature basis.

11 Baselines and Ablations

11.1 Baselines

The minimal strong baseline set is:

- random intervention sites;
- activation norm;
- gradient saliency;
- logit margin and entropy;
- attribution patching;
- activation patching sweep as an oracle;
- path patching where feasible;
- causal tracing for factual recall;
- ACDC or edge-attribution patching where feasible;
- SAE feature magnitude and SAE steering in feature experiments;
- Fisher norm;
- intrinsic dimension and persistent homology as diagnostic-only baselines;
- EICS-style sheaf inconsistency score where implementation is feasible.

11.2 Ablations

Essential ablations:

- no candidate-specific sheaf;
- identity transports;
- random transports;
- learned transports versus Jacobian transports;
- scalar logit-lens germs versus low-rank residual germs;
- residual features versus SAE features;
- obstruction without response;
- response without obstruction;
- no Fisher/KL penalty;
- different edge weights;
- different site granularities;
- paraphrase transfer;
- cross-model transfer.

12 Expected Figures and Tables

Figure 1: Local-to-global decision schematic. A Transformer computation graph with local evidence at tokens, layers, heads, MLPs, residual streams, and features. Compatible evidence glues into stable commitment; incompatible evidence produces obstruction.

Figure 2: Mathematical object diagram. Show K_x, stalks $F_c(v)$, germs $s_c(v)$, maps $\rho^c_{u\to v}$, residuals r_e, global section space H^0, and obstruction distance.

Figure 3: Obstruction-response loop.

$$\text{local evidence} \to \mathcal{O}_c \to S_u \to \text{intervention} \to \Delta\mathcal{O}_c \to \Delta m_c \to \text{repair/flip/stabilize}.$$

Figure 4: Predicted versus observed margin change. A scatter plot across sites, prompts, and tasks showing $\widehat{\Delta m_c}$ or S_u versus observed Δm_c.

Figure 5: Obstruction reduction versus repair probability. Bucket interventions by actual $-\Delta\mathcal{O}_c$ and show monotonic repair probability.

Figure 6: Baseline comparison. Top-k repair rate or AUC against attribution patching, activation norm, gradient saliency, random, and oracle activation patching.

Figure 7: Failure cases. Examples where obstruction decreases without margin movement, margin moves without obstruction reduction, or transport maps fail.

13 Hierarchy of Ideas

Table 2: Hierarchy of ideas in the paper.

Idea	Role
Local-to-global decision theory	Central spine
Obstruction-response functional	Central spine and main object
Sheaf/compatibility structure	Major mathematical section
Corrected obstruction energy	Major mathematical section
Causal response	Major algorithmic/intervention section
Activation patching	Empirical intervention tool and baseline
Causal abstraction	Related work and conceptual support
Sparse autoencoders	Major optional feature-space experiment
Information geometry	Subsection for Fisher/KL intervention cost
Decision singularities	Subsection/appendix under local boundary assumptions
Field theory	Supporting response-calculus language only
Topology	Optional diagnostic/appendix
Algebraic/tropical geometry	Appendix only under frozen-attention or local-linear assumptions
Stochastic thermodynamics	Future work unless directly tested
Hallucination	Failure-mode experiment, not central theorem
Theorem proving	Future extension or brief experiment
Agents/search	Future paper material
Renormalization/coarse-graining	Future extension unless used for site compression

14 Title Options

1. **Causal Obstruction–Response Theory of Transformer Decisions**
2. Local-to-Global Decision Mechanics in Transformers
3. Obstruction-Guided Decision Mechanics in AI Systems
4. Decision Coherence in Transformers: A Local-to-Global Theory
5. Local Evidence, Global Commitment: A Theory of Transformer Decisions
6. Obstruction, Response, and Repair in Transformer Decision-Making
7. Causal Decision Sheaves for Transformer Interpretability
8. Measuring and Repairing Local-to-Global Inconsistency in Language Models
9. The Mechanics of Transformer Decisions

10. From Local Evidence to Global Commitment in AI Systems
11. Decision Singularity Surgery in Transformers
12. A Local-to-Global Calculus for AI Decisions
13. Obstruction Energy Predicts Causal Decision Repair in Transformers
14. Transformer Decisions as Distributed Consistency Events
15. Toward a Science of AI Decision Mechanics

The recommended title is:

Causal Obstruction–Response Theory of Transformer Decisions

15 Novelty and Safety

15.1 Unsafe Claims

Avoid:

- "We introduce the first use of sheaf theory in Transformers."
- "We introduce the first local-to-global theory of neural networks."
- "Hallucination is obstruction."
- "We prove AI decisions are global sections."
- "Field theory of Transformer patching is new."
- "We identify the true circuit."
- "Wrong answers are exactly failures of gluing."
- "This is a universal theory of intelligence."

15.2 Safe Claims

Use:

- "We propose a candidate-specific local-to-global framework for Transformer decisions."
- "We define obstruction as distance to global compatibility, computed by a Laplacian residual."
- "We test whether obstruction-response rankings predict causal intervention effects."
- "The method identifies intervention-effective structures, not necessarily unique circuits."
- "Obstruction complements confidence, entropy, and margin by measuring internal incompatibility."
- "Local-to-global consistency is a modeling principle, not a universal law."

16 Reviewer-Proof Positioning

1. **This is just activation patching with extra math.** Activation patching measures effects by intervention; obstruction-response predicts which interventions should work and why.
2. **The sheaf is arbitrary.** The paper includes identity, random, learned, and Jacobian transport ablations.
3. **Transport maps are post hoc.** Transports are trained on clean examples and tested out-of-distribution on corrupt, ambiguous, and paraphrased examples.
4. **Obstruction only tracks margin.** Report partial correlations and predictive lift over margin and entropy.
5. **This does not prove decisions are global sections.** The paper does not claim literal global-section cognition; it tests a compatibility model.
6. **SAE bases are unstable.** Compare residual, PCA, probe, and SAE bases.
7. **Universality is overclaimed.** Universal language is restricted to discussion; main claims are empirical and candidate-specific.
8. **Field theory and topology are decorative.** They are not central; response theory and obstruction are central.
9. **Related work already uses sheaves.** The novelty is candidate-specific obstruction-response and causal intervention prediction.
10. **Interventions do not transfer.** Transfer is explicitly measured; lack of transfer becomes a limitation.
11. **Attribution patching is stronger.** Attribution patching is a primary baseline.
12. **Patching conclusions depend on metric choice.** The paper reports multiple metrics: margin, KL, repair rate, flip probability, and obstruction change.
13. **Non-unique circuits undermine the framework.** The framework claims intervention-effective structures, not unique mechanisms.
14. **Obstruction reduction may be epiphenomenal.** The paper tests mediation by measuring whether $-\Delta\mathcal{O}_c$ predicts Δm_c after controls.
15. **Math is too heavy.** Only the obstruction definition, energy-distance theorem, and response rule are central.

17 Minimum Viable, Strong, and Landmark Versions

17.1 Minimum Viable Paper

Scope: GPT-2 small and Pythia-small models; IOI and one factual recall task.

Theorem set: energy-distance equivalence, minimum-energy repair, and cohomology correction.

Claims: obstruction-response predicts causal repair sites better than random, activation norm, gradient saliency, and basic attribution patching on at least one known-circuit task.

17.2 Strong Conference Paper

Scope: GPT-2, Pythia, and one modern open-weight model; IOI, factual recall, MCQA, and ambiguity.

Theorem set: energy-distance equivalence, repair rule, obstruction-margin stability, non-identifiability, local boundary theorem in appendix.

Claims: obstruction-response generalizes across tasks and adds predictive value beyond strong patching and saliency baselines.

17.3 Landmark Paper

Scope: multiple models, multiple tasks, residual and feature-space implementations, robust transfer, public benchmark.

Theorem set: all main results plus empirically validated linear-response regimes.

Claims: local-to-global obstruction is a causal predictor of decision repair and stability, providing a new way to study AI decision mechanics.

The unforgettable figure would show predicted obstruction-response score versus observed causal margin change across tasks and models, with clear superiority over attribution patching and other baselines.

18 Conclusion

The exact paper goal is to make local-to-global consistency a falsifiable theory of Transformer decision formation.

The exact central contribution is the obstruction-response functional: a corrected obstruction energy coupled to a causal intervention rule.

The center of the paper should be the loop

$$\text{local evidence} \to \text{obstruction} \to \text{predicted intervention} \to \Delta\text{obstruction} \to \Delta\text{margin} \to \text{repair/flip/stabilization}.$$

Supporting material should include sheaf formalism, energy-distance equivalence, Fisher/KL intervention cost, decision-boundary analysis, and carefully scoped failure-mode experiments.

Cut or demote anything that does not improve this loop: decorative field theory, broad thermodynamics, speculative topology, unrestricted tropical geometry, and premature claims about agents or theorem proving.

The sentence to keep in mind while writing the paper is:

> *A theory of AI decisions becomes scientific only when its internal invariant predicts a causal intervention.*

References

[1] Nanda, N. and Bloom, J. TransformerLens: A Library for Mechanistic Interpretability of Generative Language Models. Software documentation and open-source library.

[2] Meng, K., Bau, D., Andonian, A., and Belinkov, Y. Locating and Editing Factual Associations in GPT. *NeurIPS*, 2022.

[3] Wang, K. R. and colleagues. Interpretability in the Wild: A Circuit for Indirect Object Identification in GPT-2 Small. 2022.

[4] Conmy, A., Mavor-Parker, A., Lynch, A., Heimersheim, S., and Garriga-Alonso, A. Towards Automated Circuit Discovery for Mechanistic Interpretability. 2023.

[5] Syed, A., Rager, C., and Conmy, A. Attribution Patching Outperforms Automated Circuit Discovery. 2023.

[6] Geiger, A., Wu, Z., Potts, C., Icard, T., and Goodman, N. Finding Alignments Between Interpretable Causal Variables and Distributed Neural Representations. 2021.

[7] Bricken, T. et al. Towards Monosemanticity: Decomposing Language Models With Dictionary Learning. 2023.

[8] Farquhar, S., Kossen, J., Kuhn, L., and Gal, Y. Detecting Hallucinations in Large Language Models Using Semantic Entropy. *Nature*, 2024.

[9] Yao, S. et al. Tree of Thoughts: Deliberate Problem Solving with Large Language Models. 2023.

[10] Zhang, L., Naitzat, G., and Lim, L.-H. Tropical Geometry of Deep Neural Networks. *ICML*, 2018.

[11] Placeholder. Neural Networks as Local-to-Global Computations. Add final citation.

[12] Placeholder. Effective-Information Consistency Score for Transformer Circuits. Add final citation.

[13] Placeholder. Transformer Field Theory and Response Functions for Patching. Add final citation.

[14] Placeholder. Persistent Topology of Language Model Representations. Add final citation.